

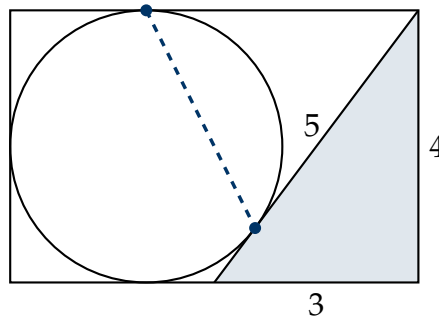
Problem of the Week

Due Monday, October 12 at 11:59 pm (Arizona time)

Two Points of Contact

A circle and a right triangle with sides 3, 4, 5 sit inside a rectangle. The circle touches the top, bottom, and left sides of the rectangle. The triangle sits in the lower-right corner: its leg of length 3 lies along the bottom side, its leg of length 4 runs the full height of the right side, and its hypotenuse is tangent to the circle.

Find the length of the segment joining the point where the circle touches the top side to the point where the circle touches the hypotenuse (the dashed segment in the figure).



How to enter

Scan the code, or go to ac.nau.edu/lms-apps/self-enroll/545026, to join the POTW Canvas course. Then upload **one PDF** (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

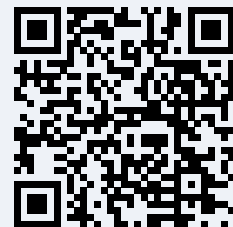
The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, ...), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:

naumathstat.github.io/problem-of-the-week

Questions: ian.williams@nau.edu



Join on Canvas